

Probability and Statistics – for Computer Science

Based on Forsyth, *Probability and Statistics for Computer Science* (2018)

Applied Statistics and Probability for Computer Scientists

Basic Ideas in Probability

Chapter 3 — Experiments, Events, Independence, and Conditioning

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Chapter Outline | What we will cover

Experiments & Sample Spaces
Events & Counting
Independence

Conditional Probability
Famous Paradoxes
Summary

Learning Objectives

- ▶ Build **sample spaces** and compute probabilities by counting
- ▶ Manipulate **events** with set operations and combinatorics
- ▶ Test for **independence** and avoid the gambler's fallacy
- ▶ Apply **conditional probability** and **Bayes' rule**



Section 1

Experiments & Sample Spaces

What Is an Experiment? | A repeatable procedure with an uncertain outcome

Definition 3.1: Sample space

The **sample space** Ω is the set of *all* possible outcomes of an experiment.

- ▶ Flip a coin: $\Omega = \{H, T\}$
- ▶ Roll a die: $\Omega = \{1, 2, 3, 4, 5, 6\}$

Worked Example: Find the Lady

Three cards, one queen, shuffled face down.
Outcome = which position hides the queen.

$$\Omega = \{1, 2, 3\}$$

Probability of Outcomes | Assigning weights that must add to one

Formal definition

A probability assigns $P(a) \geq 0$ to each $a \in \Omega$ with

$$\sum_{a \in \Omega} P(a) = 1$$

A biased coin

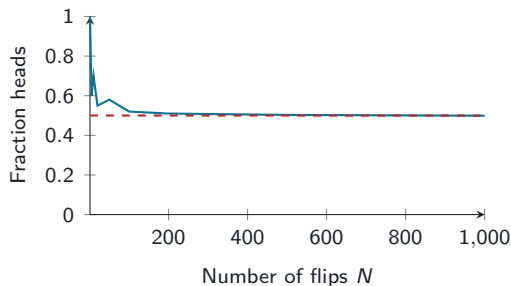
$P(H) = 0.7$, $P(T) = 0.3$ is a valid assignment — outcomes need not be equally likely.

Equally likely outcomes

A **fair** die has $P(i) = 1/6$ for each face.

The Law of Large Numbers, in Action | Long-run frequency converges to probability

As the number of trials N grows, the observed frequency of an outcome converges to its true probability $P(a)$.



Computing Probabilities by Counting | Two dice: 36 equally likely pairs

Equally likely outcomes

If all $|\Omega|$ outcomes are equally likely,

$$P(a) = \frac{1}{|\Omega|}$$

Two fair dice

$|\Omega| = 6 \times 6 = 36$ ordered pairs.

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

(6 pairs sum to 7: $(1, 6), (2, 5), \dots, (6, 1)$)

Section 2

Events & Counting

Definition 3.2: Event

An **event** E is any subset of Ω ; it "occurs" if the outcome lies in E .

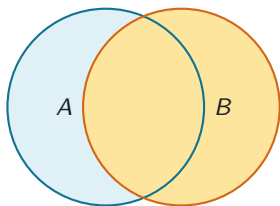
Odd numbers, fair die

$$E = \{1, 3, 5\}, P(E) = 3/6 = 1/2$$

Basic properties

- ▶ $0 \leq P(E) \leq 1$
- ▶ $P(\Omega) = 1$
- ▶ Disjoint E_1, E_2 :
 $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

Set Operations on Events | The size analogy



General addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Complement

$$P(A^c) = 1 - P(A)$$

Permutations & Combinations | Counting without listing

Concept	Meaning	Formula
Permutation	Ordered arrangement of k from n	$\frac{n!}{(n-k)!}$
Combination	Unordered choice of k from n	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Card hands

Ordered 5-card hands: $52 \cdot 51 \cdot 50 \cdot 49 \cdot 48$. Unordered (poker) hands: $\binom{52}{5} = 2,598,960$.

The Birthday Problem | A famously counter-intuitive result

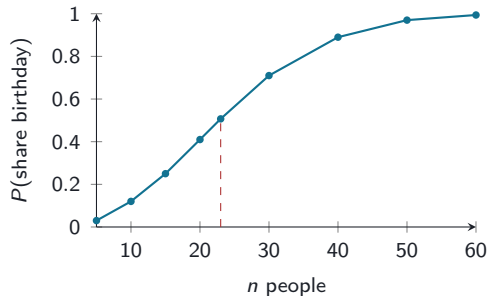
Question

With n people, what is the probability that *at least two* share a birthday?

Trick: use the complement

$P(\text{share}) = 1 - P(\text{all different})$

$$P(\text{all different}) = \prod_{i=0}^{n-1} \frac{365 - i}{365}$$



Only 23 people needed to pass 50%!

Section 3

Independence

Independent Events | Knowing one tells you nothing about the other

Definition 3.3

A and B are **independent** iff

$$P(A \cap B) = P(A) P(B)$$

Fair dice

Roll two dice. "First die is 6" and "second die is 6" are independent: $P(\text{both}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$.

The Gambler's Fallacy | Independent trials have no memory

The fallacy

"The coin landed heads five times in a row — tails is *due* now." False: each flip is independent.

The truth

$P(\text{tails on flip 6}) = 0.5$ regardless of past flips — the coin has no memory.

Case Study: Airline Overbooking | Using independence to model no-shows

Setup

A flight sells n tickets for c seats. Each passenger independently shows up with probability p .

Oversold probability

Number who show up $\sim \text{Binomial}(n, p)$.

$$P(\text{oversold}) = P(X > c)$$

Airlines pick n so this stays acceptably small.

Section 4

Conditional Probability

Definition 3.4

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

The **size metaphor**: restrict the sample space to B , then ask what fraction of it is also in A .

Non-independent events

Draw one card. A = "face card", B = "heart".

$$P(A | B) = \frac{3/52}{13/52} = \frac{3}{13} \neq P(A)$$

Three Key Formulas | Multiplication rule, total probability, Bayes' rule

Rule	Formula
Multiplication	$P(A \cap B) = P(A B) P(B)$
Total probability	$P(B) = \sum_i P(B A_i) P(A_i)$ for a partition $\{A_i\}$
Bayes' rule	$P(A B) = \frac{P(B A) P(A)}{P(B)}$

Worked Example: Car Factories | Applying total probability and Bayes' rule

Factory A makes 60% of cars, 2% defective.
Factory B makes 40%, 5% defective. A random car is defective — which factory made it?

Solution

$$P(D) = 0.6(0.02) + 0.4(0.05) = 0.032$$

$$P(A | D) = \frac{0.6 \cdot 0.02}{0.032} = 0.375$$

Detecting Rare Events Is Hard | Why "99% accurate" is not the same as "99% reliable"

Setup

Disease prevalence 1%. Test sensitivity 99%, specificity 99%. You test positive — what is $P(\text{disease} \mid +)$?

Bayes' rule at work

$$P(+) = 0.99(0.01) + 0.01(0.99) = 0.0198$$

$$P(D \mid +) = \frac{0.99 \cdot 0.01}{0.0198} \approx 0.50$$

Key insight

When the base rate is low, most positives are **false positives** — even with a highly accurate test.



Section 5

Famous Paradoxes

Warning Example: The Prosecutor's Fallacy | Confusing $P(E \mid \text{innocent})$ with $P(\text{innocent} \mid E)$

The Sally Clark case

A rare event (two infant deaths) was presented as strong evidence of guilt using only $P(E \mid \text{innocent})$, ignoring the **base rate** of guilt vs. innocence.

The correct question

What matters is $P(\text{innocent} \mid E)$, obtained via Bayes' rule — *not* the reverse conditional.

The Monty Hall Problem | Switch or stay?

The game

3 doors, 1 car. You pick a door; the host, who knows where the car is, opens *another* door with a goat. Should you switch?

The answer depends on the host's rule

If the host *always* opens a goat door:

Strategy $P(\text{win})$	
Stay	1/3
Switch	2/3

Section 6

Summary

Formula Sheet | The core toolkit of Chapter 3

Concept	Formula
Equally likely	$P(a) = 1/ \Omega $
Complement	$P(A^c) = 1 - P(A)$
Union	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Independence	$P(A \cap B) = P(A)P(B)$
Conditional	$P(A B) = P(A \cap B)/P(B)$
Bayes' rule	$P(A B) = P(B A)P(A)/P(B)$

Chapter Summary | Key takeaways

Model

Sample spaces, events, and probability assignments give a precise language for uncertainty.

Reason

Independence simplifies computation; conditioning updates beliefs with evidence.

Watch out for

The gambler's fallacy, the prosecutor's fallacy, and confusing $P(A | B)$ with $P(B | A)$.

Looking ahead

Ch. 4: **Random variables** turn outcomes into numbers we can average.

Thank you!

Questions & Discussion

Next: Chapter 4 — Random Variables and Expectations